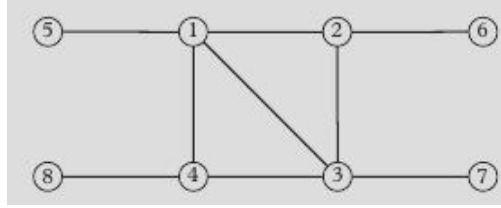


HOMEWORK III

1. Consider the network shown in the figure.



- a) The average shortest path length L for a network of N nodes is defined by

$$L = \frac{1}{N(N-1)} \sum_{i \neq j} d_{ij},$$

where d_{ij} is the length of the shortest path between node i and node j .

Calculate L for this network. [1]

- b) The local clustering coefficient of a node i is defined as

$$C(i) = \frac{2t_i}{k_i(k_i - 1)},$$

where t_i is the number of triangles containing node i and k_i is the number of neighbors of node i .

The clustering coefficient of the network is the average

$$\langle C \rangle = \frac{1}{N} \sum_{i=1}^N C(i).$$

Calculate the average clustering coefficient $\langle C \rangle$ for this network. [1]

- c) The degree or number of neighbors of node i is denoted by k_i . Calculate the average degree of this network. [1]

2. Describe the properties of the following networks and the mechanisms proposed for their formation:
 - a) Small world networks. [1]
 - b) Scale-free networks. [1]
 - c) Communities. [1]
3. a) Describe the phase transition found in the Axelrod model for cultural influence. [1]
 b) Describe the counterintuitive effect of external mass media in Axelrod model. [1]
4. Implement a computer code for Deffuant's bounded confidence model in a globally coupled network of N agents with approaching parameter μ and interaction threshold d (see Class Notes). You may use the programming language of your choice. Plot the temporal evolution of the opinion states of the $N = 500$ agents until clusters are formed, with fixed parameters:
 - a) $\mu = 0.5, d = 0.1$. [1]
 - b) $\mu = 0.5, d = 0.4$. [1]